

UNIVERSITY COMMITTEE EDITION

MedLine

Medication delivery and medical logistics across patients, pharmacies, warehouses, drivers and administrators.

30 minutes	4 speakers	5 operational roles
Focused committee presentation	Speaker 1, 2, 3 and 4	One server-controlled workflow

Prepared for live demonstration and technical discussion

Current solution baseline: 24 August 2026

Executive summary

MedLine connects the complete medicine journey through role-specific interfaces and a single authoritative workflow engine.

100 seeded medicines	10 Damascus pharmacies	2 warehouses	2 rated drivers
--------------------------------	----------------------------------	------------------------	---------------------------

What the solution demonstrates

- A patient can discover an approved pharmacy, select medicines, attach item-specific prescriptions, choose a road-routed delivery, and follow the result.
- A pharmacy can review prescriptions and quantities, manage inventory and working hours, replenish from warehouses, and control pickup handoff verification.
- A warehouse can maintain traceable batches, allocate exact quantities, approve full or partial procurement, and dispatch through the same delivery network.
- A driver can review vehicle-compatible jobs on a route map, accept an order, receive secure pickup, mark arrival, and complete recipient verification.
- An administrator governs organizations, subscriptions, users, catalog, pricing, complaints, ratings, audit records, delivery recovery and operational health.

Central design decision

Laravel is the authority for permissions, status transitions, prices, reservations, handoff verification and ownership. React and Flutter render role-aware experiences but cannot invent business state.

The 30-minute committee run of show

The presentation is divided evenly across four speakers and ends with a short synthesis and questions.

Time	Owner	Presentation responsibility	Primary live evidence
00:00-07:00	Speaker 1	Problem, solution thesis, stakeholder roles and architecture	Opening slides and role map
07:00-14:00	Speaker 2	Patient order, pharmacy review, inventory and warehouse procurement	Order queue, inventory and partial offer
14:00-21:30	Speaker 3	Driver map, road pricing, pickup verification, transit, arrival and recipient handoff	Route map and delivery detail
21:30-30:00	Speaker 4	Administration, assurance, theater data, release boundaries and closing	Admin controls, audit and demo summary

Handoff discipline

- Each speaker ends by creating the question that the next speaker answers.
- Do not read status tables aloud. Explain one concrete scenario, then use the tables only when the committee asks for detail.
- Use the presentation screenshots if the network or emulator is slow; use the live portal only for the planned click path.
- Keep 60-90 seconds inside the final slide for committee questions without exceeding 30 minutes.

One sentence every speaker should repeat

The client shows the workflow; the API enforces it.

Solution scope and role model

Five user roles see different dashboards and actions, while support remains a seeded operational account rather than a committee demo persona.

Capability	Patient	Pharmacy	Warehouse	Driver	Admin
Catalog	Browse	Browse	Browse	Manifest	Manage
Patient orders	Create / decide	Review / fulfil	-	Deliver	Monitor
Procurement	-	Create / decide	Review / fulfil	Deliver	Monitor
Inventory	-	Own stock	Own batches	-	Oversight
Delivery	Track	Pickup / monitor	Pickup / monitor	Accept / progress	Reassign
Subscription	None	Own plan	Own plan	None	Review
Governance	Own support	Own support	Own support	Own support	Users / audit / policy

Ownership rules

- Patients see their orders, addresses, prescriptions, deliveries, complaints and ratings.
- Pharmacies and warehouses see only records connected to their approved partner profile and active subscription.
- Drivers see available jobs matching their approved vehicle type plus their own assignments.
- The administrator is deliberately not associated with any pharmacy or warehouse.
- Navigation visibility improves usability; server-side role middleware and ownership checks provide security.

Architecture and trust boundaries

The system is a native Windows multi-client platform with replaceable provider integrations and transaction-safe MySQL persistence.

Layer	Implemented technology	Responsibility
Web portal	React + TypeScript + Vite	Role dashboards, operational queues, forms, maps and administration
Mobile	Flutter with Android flavors	Role-based mobile parity, secure session, maps, files and driver workflow
API	Laravel 12 + Sanctum	Authorization, validation, workflows, pricing, transactions and signed file access
Data	MySQL 8 + private storage	Accounts, stock, orders, events, audit and protected documents
Realtime / jobs	Laravel Reverb, queue worker and scheduler	Live updates, external notification dispatch and maintenance
Providers	OpenStreetMap/routing, SMTP, FCM and optional SMS	Road routes, email, push and replaceable external channels

Consistency boundary

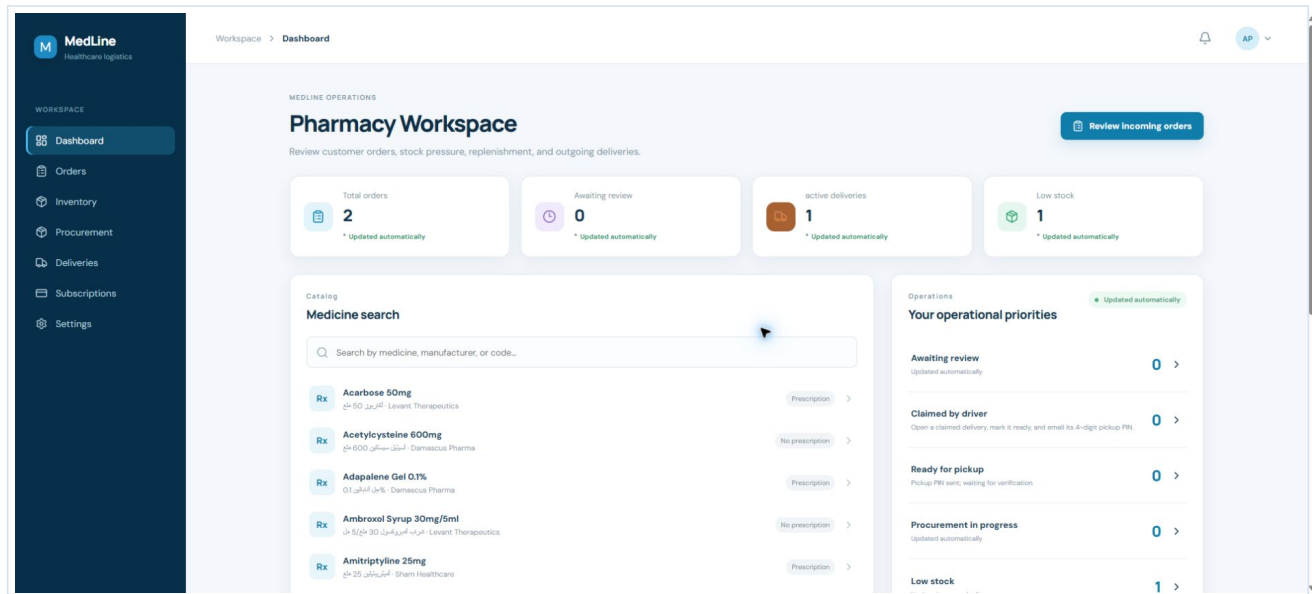
Critical stock reservations, delivery claims, pickup verification, recipient completion, payment settlement and inventory movements run inside bounded database transactions with row locks and retry-safe idempotency.

Deployment model

- No Docker. Laravel, React, MySQL, Reverb, workers, scheduler and Flutter run as native services or processes.
- Production secrets, TLS, provider accounts and signing keys remain outside source control.

Role-aware workspaces make priorities visible

Dashboards adapt to the signed-in role and lead directly to the work that user can perform.



Live local pharmacy workspace: order volume, review queue, deliveries, low stock, bilingual catalog and operational priorities.

- Patient dashboards emphasize discovery, orders and active delivery.
- Pharmacy dashboards emphasize incoming orders, low stock, procurement and pickup readiness.
- Warehouse dashboards emphasize fulfilment, batch inventory and dispatch.
- Driver dashboards emphasize available route-compatible work and active assignments.
- Administrator dashboards emphasize system-wide health, approvals and intervention queues.

Patient experience

The patient journey combines catalog discovery, controlled prescriptions, transparent pricing and secure fulfilment.

Discovery and account

- English/Arabic medicine search by name, code, manufacturer and form.
- Approved pharmacy discovery with address, map pin and working hours.
- Saved map-pinned addresses, profile, locale, password recovery, consents and notification preferences.

Order creation

- Multiple medicines and quantities in one order.
- One private prescription per prescription-required medicine.
- ASAP or scheduled delivery, vehicle choice and readable destination snapshot.
- Road distance, rate per kilometre, delivery fee, tax and total shown before submission.

After submission

- Prescription and pharmacy review timeline.
- Accept or reject a pharmacy partial offer.
- Follow the related delivery without duplicating delivery state in the order.
- Submit a 1-5 rating and open a complaint with optional private evidence.

Important distinction

An accepted order is not automatically completed. The order becomes completed only when the final recipient handoff is verified and inventory/payment settlement succeeds.

Pharmacy operations

A pharmacy is both a clinical review point and a logistics participant.

Workspace > Orders

Orders overview 2 orders

Search, review and manage every order from one place.

Search orders

Order, customer, medicine or destination

Status: All statuses

ORDER	CUSTOMER	PHARMACY	DRIVER	MEDICINES	DESTINATION	TOTAL	STATUS	CREATED	ACTION
THR-ORD-COMLETE2-D2	Demo Patient One	Al-Razi Pharmacy	Demo Driver Two	Clopidogrel 75mg	12 Faysz Mansour Street, Al-Mazzez, Damascus	SYP 2,374.5	Completed	02 Aug 2026 16:49	
THR-ORD-PARTIAL-ACCEPT	Demo Patient Two	Al-Razi Pharmacy	Demo Driver One	Rosuvastatin 10mg	18 Bab Touma Street, Bab Touma, Damascus	SYP 3,601	Partially Accepted	24 Jul 2026 16:49	

Rows per page: 10

Previous Page 1 of 1 Next

The queue exposes customer, pharmacy, driver, medicines, destination, total, status and creation time without losing record-level detail.

- Open and review each medicine and its prescription independently.
- Approve or reject a prescription; rejection requires a patient-facing note.
- Accept the full order, reject it, or reduce quantities and submit a partial offer.
- A partial offer cannot be submitted unless at least one requested quantity is actually reduced.
- Manage own inventory, low-stock thresholds, working hours, procurement, subscription and related deliveries.
- After a driver claims a delivery, send the four-digit pickup PIN and verify the code shown by the driver.

Warehouse, inventory and procurement

The stock model preserves batch traceability and prevents fulfilment decisions from exceeding real inventory.

MEDICINE	OWNER	AVAILABLE	RESERVED	UNIT PRICE	BATCH / EXPIRY	STOCK HEALTH	ACTION
Acarbose 50mg	Al-Razi Pharmacy	70	0	SYP 2,970.00	PH05-038 Expires 23 Feb 2028	Healthy	
Acetylcysteine 600mg	Al-Razi Pharmacy	133	0	SYP 1,320.00	PH05-047 Expires 23 Feb 2028	Healthy	
Ambroxol Syrup 30mg/5ml	Al-Razi Pharmacy	50	0	SYP 1,470.00	PH05-048 Expires 23 Feb 2028	Healthy	
Antacid Suspension	Al-Razi Pharmacy	92	0	SYP 2,370.00	PH05-054 Expires 23 Feb 2028	Healthy	
Budesonide Inhaler 200mcg	Al-Razi Pharmacy	98	0	SYP 570.00	PH05-042 Expires 23 Feb 2028	Healthy	
Clopidogrel 75mg	Al-Razi Pharmacy	103	0	SYP 1,770.00	PH05-030 Expires 23 Feb 2028	Healthy	
Dextromethorphan Syrup 15mg/5ml	Al-Razi Pharmacy	57	0	SYP 1,620.00	PH05-049 Expires 23 Feb 2028	Healthy	

Inventory records expose available and reserved quantities, price, batch/expiry data and stock health.

- Only administrators create global medicine catalog records; warehouses stock active catalog medicines as separate batches.
- A batch records lot number, manufacture date, expiry, received date, location, quantity, reserved quantity, unit price and active state.
- Pharmacies build multi-item replenishment requests against eligible warehouse stock.
- Warehouses allocate every fulfilled unit across one or more eligible batches.
- Full acceptance requires complete allocation. Partial or rejected decisions require explanation.
- Completion consumes warehouse reservations and adds inventory to the destination pharmacy in one settlement.

Patient order status dictionary

Order status describes clinical/commercial acceptance. Physical delivery progress stays on the related delivery record.

Status	Meaning	Next normal outcome
prescription_required	At least one required prescription is missing or rejected.	Upload or replace item evidence
prescription_review	All required files exist and await pharmacy review.	Review each prescription
pending_pharmacy_review	Evidence is resolved and the pharmacy must decide quantities.	Accept, partial offer or reject
partial_approval_required	The pharmacy offered fewer units and needs patient consent.	Patient approves or rejects
partially_accepted	The patient accepted the reduced fulfilment.	Create/continue delivery
partial_offer_rejected	The patient declined reduced quantities.	Terminal
accepted	The full pharmacy offer is approved; a delivery may exist.	Delivery settlement
rejected	The pharmacy declined the order.	Terminal
cancelled	The patient/admin cancelled before disallowed dispatch states.	Terminal
completed	Recipient verification, inventory movement and payment settlement succeeded.	Rating/support

Presentation shortcut

Use one partial-offer example: requested 3 units, pharmacy can supply 2, patient approves, delivery is created for 2. This single scenario demonstrates quantity bounds, consent and preserved history.

Procurement workflow and status dictionary

Procurement mirrors patient ordering but adds exact warehouse batch allocation and pharmacy acceptance of partial fulfilment.

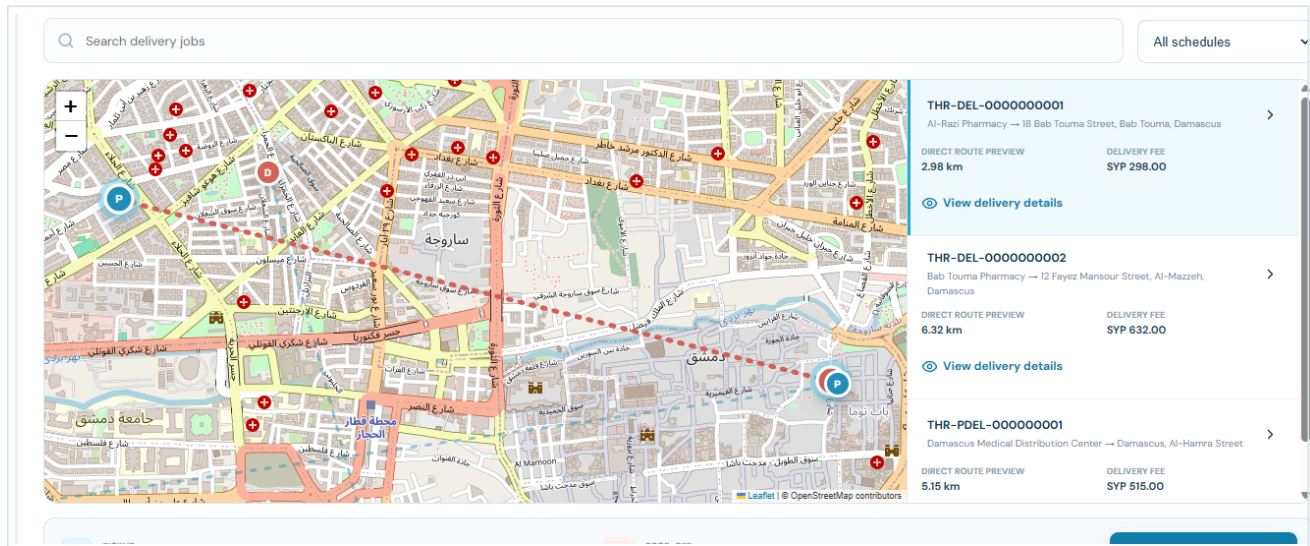
Status	Meaning	Inventory consequence
pending_warehouse_review	The warehouse must review requested medicines and quantities.	Requested stock remains reservation-controlled
partial_approval_required	The warehouse offered fewer units and awaits pharmacy consent.	Accepted quantity remains reserved
partially_accepted	The pharmacy accepted the reduced warehouse offer.	Delivery can proceed for accepted units
partial_offer_rejected	The pharmacy declined the reduced offer.	Reservations are released
accepted	The warehouse fully accepted and allocated the request.	Exact batches remain reserved
rejected	The warehouse cannot fulfil the request.	Reservations are released
completed	The delivery handoff completed successfully.	Warehouse decremented; pharmacy incremented

Procurement integrity rules

- Accepted quantity is always between zero and the requested quantity.
- A full acceptance is invalid unless every requested unit is allocated.
- A partial acceptance is invalid unless at least one quantity is reduced and a note is supplied.
- An inactive or expired batch cannot be selected for a new request.
- Pricing, schedule, route and destination are permanently captured for the request.

Driver job discovery is based on the road route

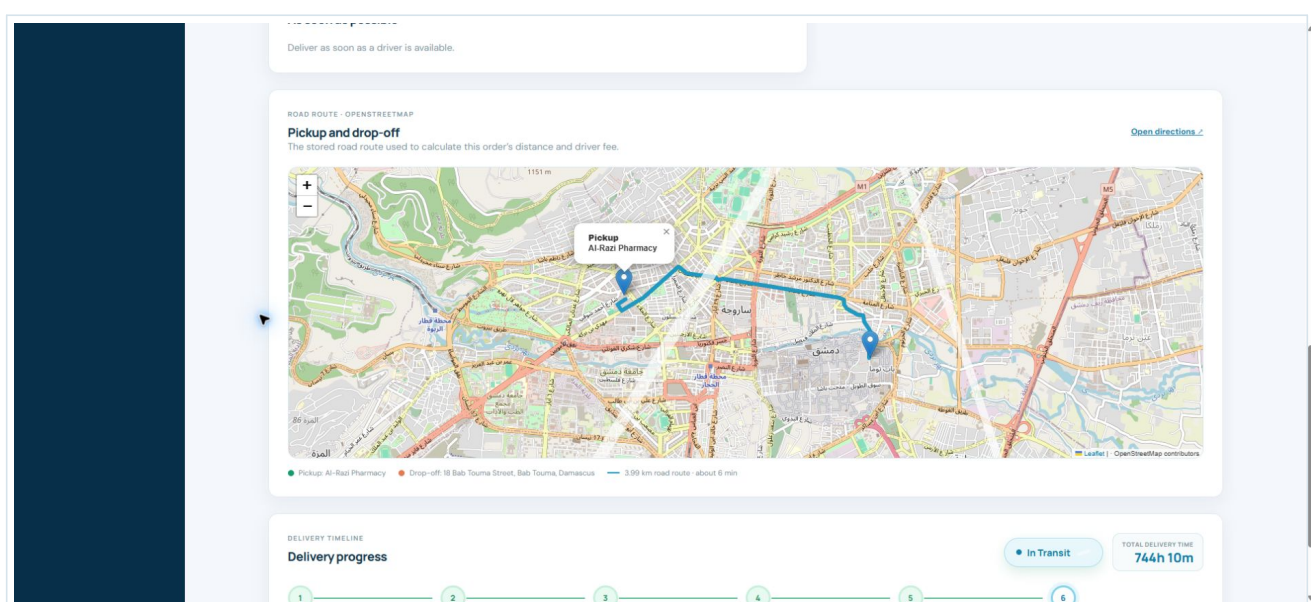
Drivers inspect the real stored route, vehicle requirement, medicine manifest, schedule and route-based fee before accepting an order.



- Available work is filtered to the driver's approved vehicle type.
- The distance and fee use the stored driving route, not straight-line distance.
- Every order snapshots the applicable rate so later administrator changes do not rewrite history.
- The driver opens full detail before acceptance: pickup, destination, recipient, contact, manifest, quantities and event timeline.
- Acceptance is serialized under a database lock so two drivers cannot claim the same job.

Road routing, pricing and recipient context

One delivery detail brings together the route evidence and the information the assigned driver needs to complete the task.



Stored OpenStreetMap road route between pickup and drop-off; the captured distance is used for the driver fee.

The screenshot shows a delivery manifest and recipient details form. The "PICKUP MANIFEST" section lists "Medicines to collect" with one item: "Rosuvastatin 10mg" (10mg Tablet, Shem Healthcare) in 2 units. The "ASSIGNMENT" section shows "THR-ORD-PARTIAL-ACCEPT" with a price of SYP 399.00. The "Delivery recipient" section shows "Demo Patient Two" with contact information: "patient.two@medline.local" and "+96391100002". The delivery destination is "18 Bab Touma Street, Bab Touma, Damascus". The "Your delivery fee" is SYP 399.00.

Driver detail includes the medicine manifest, price snapshot, recipient name, email, phone and delivery destination.

Secure delivery workflow

Two independent four-digit checks protect the pharmacy/warehouse pickup and the final recipient handoff.

Step	Actor	System action	Result
1. Accept	Driver	Accepts one eligible available order under a lock.	claimed
2. Send pickup PIN	Pharmacy or warehouse	Emails a fresh four-digit code to the assigned driver.	pickup_started
3. Verify pickup	Pickup partner	Enters the code shown by the driver after physical handoff.	in_transit automatically
4. Mark arrival	Driver	Confirms arrival at the destination.	arrived
5. Send recipient PIN	Driver	Emails a separate code to the patient or destination pharmacy.	arrived, awaiting proof
6. Verify recipient	Driver	Enters the recipient's code after in-person confirmation.	delivered and settled

Current workflow change

There is no separate driver 'Start delivery' step. Successful pickup verification immediately changes the delivery from pickup_started to in_transit. The driver's next progress action is Mark arrived.

PIN controls

- Codes are stored only as password hashes and are never returned by an API.
- Codes expire, have a resend cooldown and lock after bounded incorrect attempts.
- Codes are purpose-specific: a pickup code cannot complete recipient handoff.
- The demo mail redirect sends all outgoing messages to one controlled inbox without changing account-owned addresses.

Delivery status dictionary

The main path is sequential; failure and cancellation are explicit exception paths.

Status / event	Meaning	Permitted next action
available	No driver owns the job.	Eligible driver accepts; or cancellation
claimed	An approved driver owns the delivery.	Pickup partner sends driver PIN
pickup_started	Pickup PIN was sent; physical pickup is awaiting verification.	Pickup partner verifies PIN
in_transit	Pickup was verified and transport began automatically.	Driver marks arrived
arrived	Driver reached the destination.	Send and verify recipient PIN
delivered	Recipient verification and settlement completed.	Terminal
failed	Driver reported an unsuccessful active delivery with a reason.	Administrator may reassign
cancelled	An unclaimed/eligible delivery was cancelled.	Terminal
reassigned event	Admin returned a failed job to the queue and cleared driver/PIN/location state.	Stored delivery status becomes available

Location privacy

Only the latest driver coordinate is retained for an authorized active delivery. It becomes unavailable when stale and is cleared when the job fails, is cancelled, is reassigned or is delivered.

Supporting review and operational statuses

The rest of the platform uses short, explainable review states with administrator correction loops.

Object	Statuses	Meaning
Prescription	pending_review, approved, rejected	A medicine-specific document awaits clinical review, passes, or must be replaced.
Organization	pending, correction_required, approved, rejected, suspended	Application review and ongoing access state for a pharmacy or warehouse.
Subscription	payment_under_review, correction_required, active, expiring_soon, grace, expired, rejected	Payment review and time-based operating access.
Payment proof	under_review, correction_required, approved, rejected	Private receipt evidence and its decision.
Verification document	under_review, correction_required, approved, rejected	Identity/license evidence and its decision.
Complaint	open, in_review, resolved, rejected	Support intake, active handling and terminal outcome.
User	active, suspended	Authentication and platform access state.
Rating	visible, hidden	Submitted feedback; moderation is reversible and audited.
Inventory health	healthy, low_stock, expired, inactive	Calculated presentation of quantity, threshold, expiry and active state.
Batch allocation	reserved, released, consumed	Warehouse units held, returned or finalized by procurement.

Notifications and professional email

Business mutations commit first; user-facing notification records and external delivery follow without risking the transaction.

Channel	Behavior	Demo / production boundary
In-app	Persisted notification inbox with preferences, read and delete actions.	Always available when enabled
Email	Laravel mail transport with branded, purpose-specific messages and action buttons.	Demo redirect can force all recipients to one inbox
Push	FCM adapter uses active encrypted device tokens and records attempts.	Requires provider credentials
SMS	Optional generic HTTP adapter.	Requires an approved provider
Realtime	Reverb supports live client updates while normal reads remain available.	Defaults to port 8090 locally

Notification safety rules

- No PIN, password, access token, prescription content, storage path or payment evidence enters a notification payload.
- External provider failures do not roll back a successful order, procurement, review or delivery transaction.
- Delivery attempts are retried at the queue/provider boundary and invalid push tokens are revoked.
- For the committee demo, configure MAIL_DEMO_TO_ADDRESS to one controlled test inbox and keep QUEUE_CONNECTION=sync for the simplest email demonstration.

Security, privacy and audit

MedLine protects workflow authority and sensitive documents at the server boundary.

Control	Implemented behavior
Authentication	Sanctum access tokens, rotated single-use refresh tokens, secure browser cookie transport and native body transport.
Administrator 2FA	Six-digit TOTP is required at sign-in when authenticator protection is enabled.
Authorization	Role middleware plus record ownership checks on every protected route.
Private files	Prescriptions, proofs, identity documents and complaint evidence stay outside public storage and use short-lived signed access.
Upload safety	Type/size validation and a fail-closed scanner integration when enabled.
Mutation safety	Named throttles, idempotency keys, database transactions, row locks and bounded deadlock retries.
Auditability	Administrator decisions, downloads, pricing changes, delivery recovery and critical workflow actions are recorded.
Privacy	Latest-location model, no sensitive analytics/log payloads, consent records and owner-defined retention gates.

Production release gate

The implementation is security-conscious, but real deployment still requires owner-approved prescription policy, privacy terms, retention periods, HTTPS, provider secrets, private-storage ACLs and pilot validation.

Consistency and failure recovery

The solution is designed so visible states agree with stock, delivery events, payment status and authorization.

Database guarantees

- Stock and batch rows are locked before reservation or consumption.
- Critical operations are transactional and use bounded retry on deadlocks.
- Idempotency keys replay the same response for a retried identical request and reject key reuse with different input.

Delivery consistency

- A delivered patient order must be completed and paid.
- A delivered procurement must be completed with consumed batch allocation and destination inventory movement.
- A failed delivery clears location and may only be returned to available through the audited admin reassignment action.

Operational recovery

- Health and readiness endpoints separate process liveness from dependency readiness.
- Queues and notification delivery attempts are observable.
- Backups are native, encrypted and intended for isolated restore validation.

Why the status dictionary matters

A status is not cosmetic text. It determines who may act, what must already be true, which records move together and whether an operation is recoverable.

Web, mobile, localization and accessibility

The two clients share the same server contract while adapting interaction patterns to desktop operations and mobile field work.

Area	Web portal	Flutter mobile
Navigation	Responsive role sidebar, searchable queues and whole-row record access.	Role-gated shell, Material 3 navigation and touch-first actions.
Data lists	Search, status filters, sorting, pagination and wide-table scroll regions.	Reusable searchable/filterable/sortable/paginated record lists and pull-to-refresh.
Maps	OpenStreetMap pharmacy availability, pickup/drop-off and driver routes.	Map pins, route views, addresses and driver location permission.
Files	Authorized upload/download controls and signed private links.	Styled file selection for prescription, proof, verification and complaint evidence.
Localization	English/Arabic with LTR/RTL layout and persisted profile choice.	Persisted locale and localized role/workflow labels.
Accessibility	Semantic buttons, labels, live regions, focus states and keyboard use.	Semantic containers, tooltips, touch targets and consistent status mapping.

Parity principle

Clients may differ visually, but neither client may expose a transition that the API does not permit.

Deterministic theater dataset

The destructive non-production seeder creates a complete, internally consistent demonstration world and verifies its invariants before finishing.

18 seeded user accounts	19 patient-order scenarios	7 procurement scenarios	8 driver ratings
-----------------------------------	--------------------------------------	-----------------------------------	----------------------------

- Exactly 100 bilingual medicines across multiple categories, manufacturers, forms and prescription rules.
- Ten approved and subscribed pharmacies across Damascus, including Central Pharmacy on Al-Hamra Street.
- Two approved and subscribed warehouses with traceable stock; one warehouse is in Damascus.
- Two approved available drivers: a motorcycle driver and a car driver, each with ratings from completed jobs.
- Two patients with Damascus addresses, plus one administrator who is not associated with any partner.
- Orders, procurements and deliveries cover every current status, including full, partial, rejected, cancelled, failed and completed outcomes.
- Subscriptions, working hours, notifications, consents, prescriptions, complaints, ratings, inventory movements and audit records are pre-populated.

Destructive by design

The seeder deletes existing application data and is blocked in production. Use it only to reset a local or explicitly authorized non-production demo environment.

Committee demo accounts

All seeded accounts use MEDLINE_SEED_PASSWORD. The default local-only password is shown here for rehearsal convenience and must never be used outside the demo environment.

Role	Email	Recommended demonstration
Administrator	admin@medline.local	Dashboard, approvals, audit, pricing and recovery
Patient	demo.patient@medline.local	Catalog, order, addresses, tracking and support
Pharmacy	pharmacy@medline.local	Central Pharmacy order/inventory/pickup workflow
Warehouse	warehouse.damascus@medline.local	Batch stock and procurement fulfilment
Driver	demo.driver@medline.local	Motorcycle-compatible map, acceptance and delivery

Local default password: ChangeMe123!
Pre-staged verification code only: 2468

- Freshly initiated pickup and recipient verifications generate a new four-digit code and email it.
- The value 2468 applies only to pre-staged pickup_started and arrived seed scenarios.
- Use the demo inbox redirect to avoid sending committee rehearsal messages to the account-owned addresses.

Setup and runtime commands

One PowerShell launcher detects whether initialization is required, starts the native services and supports explicit ports.

Normal daily startup

```
.\scripts\medline.ps1
```

Reset and load the complete theater dataset

```
.\scripts\medline.ps1 --force-install --fresh-seed
```

Backend and web only

```
.\scripts\medline.ps1 --skip-mobile
```

Choose ports non-interactively

```
.\scripts\medline.ps1 --api-port 8100 --reverb-port 8190 --web-port 3100
```

Target a specific Android emulator/device

```
.\scripts\medline.ps1 --mobile-device emulator-5554
```

- Defaults: API 8000, Reverb 8090, web 3001. Port 8080 is avoided on this workstation.
- The launcher starts MySQL, Laravel, Reverb, queue worker, scheduler, Vite and Flutter in separate windows when needed.
- Use --install-only for setup/migrations without runtime processes and --select-ports for interactive port selection.

Recommended live demonstration

This route shows end-to-end capability without creating too many records during the committee session.

Minute	Speaker	Live action	Fallback
00-07	1	Use slides; briefly point to the live role-aware pharmacy dashboard.	Slide 5 screenshot
07-10	2	Open the patient order and explain item-level prescription and partial quantity handling.	Slides 5-6
10-14	2	Open pharmacy inventory, then show a procurement record with batch allocation.	Slides 7-8
14-17	3	Open the driver route map and one delivery detail before acceptance.	Slide 9 screenshot
17-21:30	3	Explain pickup PIN -> in transit -> mark arrived -> recipient PIN -> delivered.	Slides 10-12
21:30-26	4	Open admin dashboard, one subscription review and one audit record.	Slides 13-14
26-30	4	Show seed coverage, summarize assurance boundaries and invite questions.	Slides 15-16

Rehearsal rule

Do not reset the database during the live presentation. Reset and verify the dataset beforehand, keep all services running, and keep one signed-in tab per role if possible.

Speaker 1 detailed brief

Goal: establish the problem, the value proposition and the architecture without drowning the committee in implementation detail.

- Opening: 'Medicine delivery is not only transport. It coordinates prescriptions, real stock, partner approval, pricing, identity and two physical handoffs.'
- Explain why five role-specific views are required but one API must remain authoritative.
- Use the pharmacy dashboard as proof that the interface changes according to operational responsibility.
- Describe React, Flutter, Laravel, MySQL, Reverb, queues and replaceable providers at one sentence each.
- Close: 'Now that the platform boundary is clear, Speaker 2 will show how a medicine request becomes a controlled order and stock commitment.'

Likely questions for Speaker 1

- Why not one generic dashboard? Because each role owns different actions and data; server authorization mirrors that separation.
- Why Laravel as authority? Because multiple clients must agree on the same stock, status and permissions under concurrency.

Speaker 2 detailed brief

Goal: prove that MedLine handles clinical evidence, partial fulfilment and stock movement as one coherent business process.

- Start with the patient: pharmacy, medicines, item-specific prescription, address, schedule, vehicle and price review.
- Show the pharmacy queue and explain full acceptance, rejection and partial offer with bounded quantities.
- Emphasize that delivery progress is not copied into the order status; completion occurs only after delivery settlement.
- Move to procurement: the pharmacy replenishes from a warehouse and the warehouse allocates exact batches.
- Close: 'The order is approved and the stock is reserved. Speaker 3 will show how the physical medicine changes hands securely.'

Likely questions for Speaker 2

- Can accepted quantity exceed requested quantity? No; the API bounds every line.
- What prevents overselling? Reservations and quantity updates execute under database locks.

Speaker 3 detailed brief

Goal: demonstrate that driver work is route-aware and that both handoffs have independent proof.

- Open with the route map: the route is a stored road path, and the driver fee reflects kilometres travelled rather than straight-line distance.
- Explain vehicle matching and locked acceptance so the same job cannot be claimed twice.
- Pickup: partner sends the code to the driver; partner enters the code shown in person; the delivery becomes in transit immediately.
- Arrival: driver marks arrived, sends a separate code to the recipient and enters it after in-person confirmation.
- Failure: driver records a reason; admin can return only a failed delivery to available and resets sensitive state.
- Close: 'The delivery now has a complete event history. Speaker 4 will show how administrators govern and audit the whole platform.'

Likely questions for Speaker 3

- Why two codes? Pickup and recipient delivery are different trust boundaries and must not share proof.
- Where is the start-delivery step? It was removed; verified pickup now enters in_transit automatically.

Speaker 4 detailed brief

Goal: connect administration, security and deterministic demo coverage to an honest statement of readiness.

- Show role-aware admin queues: partners, subscriptions, documents, users, inventory, deliveries, complaints, ratings and audit.
- Explain correction_required as a recoverable review loop, not a rejection.
- Explain transaction safety, signed private files, hashed PINs, throttles, idempotency and auditable intervention.
- Use the theater dataset counts to prove scenario coverage and consistency checks.
- State production boundaries honestly: providers, legal policy, TLS, retention, signing and pilot acceptance still require owners.
- Close: 'MedLine demonstrates a complete medication logistics workflow whose screens, statuses and data remain consistent because one server-controlled model connects them.'

Likely questions for Speaker 4

- Is it production-certified? No claim is made; the application is implemented, while legal/provider/security release gates remain explicit.
- Can admin access everything? Admin has operational scope, but private download endpoints remain authorized and audited.

Committee question bank

Use these short answers to keep the final discussion precise and consistent across all four speakers.

Question	Concise answer
How do you prevent two drivers accepting one order?	The API locks the delivery and driver records before changing ownership.
Why use road distance?	It better represents real travel effort and permanently snapshots a reproducible fee.
Are verification codes visible in the database or API?	Only hashes are stored; values never enter API responses, logs or notification payloads.
How is stock consistency maintained?	Reservations, batch allocation and completion movements occur in transactions with row locks.
Can a pharmacy work without approval/payment?	No. Operational access requires an approved partner and active subscription.
What happens if email fails after an order succeeds?	The business transaction remains committed; provider delivery is retried separately.
Does admin belong to a pharmacy?	No. The seed and domain explicitly keep administrator accounts unassociated.
Is Arabic supported?	Yes. Web and mobile support English/Arabic, LTR/RTL and persisted locale preference.
What is not finished for production?	Legal policy, provider credentials, HTTPS/domain configuration, signing, retention approval and pilot acceptance.

Readiness and future production gates

The committee should distinguish implemented capability from owner-controlled deployment decisions.

Implemented now	Owner / production gate
Role-secured Laravel API and both clients	Production domains, HTTPS certificates and protected environment values
Order, procurement, delivery and stock workflows	Approved legal policy for prescriptions and medicine delivery
Road routes and versioned vehicle rates	Delivery zones, commercial rates, taxes and provider terms
Private files, signed URLs and scanner adapter	Storage ACLs, backup policy and production scanner
In-app/email/push/SMS abstraction	SMTP, FCM and SMS credentials plus owner provider testing
Flutter flavors and Android build handoff	Release signing, store accounts and physical-device pilot
Privacy controls and retention commands	Approved retention periods, consent text and deletion/legal-hold process

Recommended next step

Run a five-role acceptance rehearsal on the exact committee hardware, verify the demo inbox, pre-open the map routes, and record a screen-capture fallback before presentation day.

Closing synthesis

MedLine is strongest when presented as one connected control system for medicine logistics, not as separate applications.

5 role-specific experiences	2 secure handoffs	1 authoritative API	0 manual start-delivery steps
---------------------------------------	-----------------------------	-------------------------------	---

What the committee should remember

- The patient sees clarity: real pharmacies, prescriptions, pricing and tracked fulfilment.
- Partners see control: bounded decisions, traceable inventory and subscription-based access.
- Drivers see actionable work: real routes, compatible jobs, recipient context and secure proof.
- Administrators see accountability: approvals, recovery, pricing history, complaints, moderation and audit.
- The system preserves consistency because every important transition is validated and recorded by the API.

Final sentence

MedLine turns a complex medication journey into a visible, secure and auditable workflow from inventory to recipient.

Prepared for Speaker 1, Speaker 2, Speaker 3 and Speaker 4